Sheet 4

1.a)

to show $f(x) = f(gx)$ $\quad$ g is orthogonal matrix

$$f(gx) = \|gx\| = \sqrt{(gx)^T(gx)}$$
$$= \sqrt{x^T g^T g x} = \sqrt{x^T x} = \|x\| = f(x)$$

1.b)

to show $g \nabla_x f(x) = \nabla_x f(gx)$

Let $Df(x)$ be the total derivative of f at point x.

$$f(gx) = f(x)$$
$$D f(gx)\, g = D f(x)$$
$$(\nabla f(gx))^T g = (\nabla f(x))^T$$
$$g^T \nabla f(gx) = \nabla f(x)$$
$$\nabla f(gx) = g \nabla f(x) \qquad g g^T = I$$

1.c)

$$u(x) = \nabla \|x\| = \nabla \sqrt{x^T x} = \nabla \sqrt{\Sigma_i x_i^2}$$
$$= \frac{1}{2\sqrt{\Sigma_i x_i^2}}\, 2x = \frac{x}{\|x\|}$$

1. d)

$$Du(x) =$$

$i \neq j$:

$$(Ju)_{ij} = \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j} \frac{x_i}{\|x\|} = \frac{\partial}{\partial x_j} \frac{x_i}{\sqrt{\Sigma_i x_i^2}}$$
$$= x_i \frac{\partial (\Sigma_i x_i^2)^{-\frac{1}{2}}}{\partial \Sigma_i x_i^2} \frac{\partial \Sigma_i x_i^2}{\partial x_j}$$
$$= x_i \left(-\tfrac{1}{2} (\Sigma_i x_i^2)^{-\frac{3}{2}}\right) 2 x_j$$
$$= \boxed{- \frac{x_i x_j}{\|x\|^3}}$$

$i = j$:

$$(Ju)_{ii} = \frac{\partial}{\partial x_i} \frac{x_i}{\|x\|} = \frac{\partial x_i}{\partial x_i} \|x\|^{-1} + x_i \frac{\partial \|x\|^{-1}}{\partial x_i}$$
$$= \boxed{\frac{1}{\|x\|} - x_i \frac{1}{\|x\|^3}}$$

$$\frac{\partial}{\partial x_i} \frac{1}{\sqrt{\Sigma_i x_i^2}} = - \frac{1}{\|x\|^3}$$

$$\frac{\partial}{\partial x_i} \frac{1}{\|x\|} = \frac{\partial}{\partial x_i} \frac{1}{\sqrt{\Sigma_i x_i^2}} = \frac{\partial}{\partial x_i} -\frac{1}{2}(\Sigma_i x_i^2)^{-\frac{3}{2}}\, 2 x_i$$